Q: If $(A^t - 2I)^{-1} = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$ then find A

Solution: Given

$$(A^t - 2I)^{-1} = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$$

$$\left[(A^t - 2I)^{-1}\right]^{-1} = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}^{-1}$$

$$A^t - 2I = \frac{\operatorname{adj}\begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}}{\begin{vmatrix} 3 & 0 \\ 1 & -1 \end{vmatrix}}$$

(as $(A^{-1})^{-1} = A$)
and $A^{-1} = \frac{\operatorname{adj} A}{|A|}$.

$$A^t = \frac{\begin{bmatrix} -1 & 0 \\ -1 & 3 \end{bmatrix}}{3} + 2I$$

$$A^t = \frac{1}{3}\begin{bmatrix} -1 & 0 \\ -1 & 3 \end{bmatrix} + 2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^t = \begin{bmatrix} -\frac{1}{3} & 0 \\ -\frac{1}{3} & 1 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A^t = \begin{bmatrix} -\frac{1}{3}+2 & 0+0 \\ -\frac{1}{3}+0 & 1+2 \end{bmatrix}$$

$$A^t = \begin{bmatrix} \frac{5}{3} & 0 \\ -\frac{1}{3} & 3 \end{bmatrix}$$

$$(A^t)^t = \begin{bmatrix} \frac{5}{3} & 0 \\ -\frac{1}{3} & 3 \end{bmatrix}^t$$

Thus

$$A = \begin{bmatrix} \frac{5}{3} & -\frac{1}{3} \\ 0 & 3 \end{bmatrix}$$

(as $(A^t)^t = A$).